

Arjun Nanduri

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Education

University of California: Berkeley, BA in Computer Science

Expected May 2028

- GPA: 3.9 | SAT: 1590
- Courses: Probability & Stochastic Processes, Machine Learning, Computer Security, Data Structures

Projects

Scripted Survey Browser AI Agent, *CalHacks Prize Winner*

<https://devpost.com/software/potbot>

- Won sponsor prize at CalHacks — led team of 3, owned backend and CV pipeline; **shipped working SF demo in 36 hours** over an unfinished feature set.
- Built agentic pipeline using a Llama vision model to validate infrastructure damage photos and **auto-submit repair requests to SF city government** via Playwright browser automation.

DealScout — Resale Arbitrage Agent (LangGraph, Playwright)

github.com/FavouritePlayer/DealScout

- Built a LangGraph-orchestrated agent that scrapes live Craigslist listings via Playwright, scores resale value with **deterministic arithmetic kept separate from LLM judgment**, then filters and ranks candidates through a structured-output LLM call.

- Integrated HydraDB as a per-user memory store so rejection feedback persists across sessions and improves future filtering; **diagnosed a silent dependency conflict** that was breaking vector DB writes with no error at insert time.

Context Custodian — Workspace Hygiene Agent (FastAPI)

github.com/FavouritePlayer/ContextCustodian

- Built a FastAPI agent that audits and cleans an agent-first workspace, acting as the user via the Scalekit SDK; **every fix is a reversible move, never a delete**.
- Designed a three-stage prompt-injection detection pipeline (regex, vector similarity, LLM adjudication) with a fail-open fallback, and engineered fix reporting that's **honest about whether an action executed live or only updated the local index**.

Experience

Machine Learning Engineer, Aver Technologies Inc. – Virginia

Aug 2025 – Present

- Built multi-stage PyTorch CV pipeline (YOLO detection + segmentation on painted depth markings, audio-visual hit sync, depth-map distance conversion) on 1000fps footage, **automating manual DOT pile depth submission**.
- Designing two-phase roadmap: Phase 1 CV digitization of manual process, **Phase 2 LIDAR elimination of markings**.

Software Engineering Intern, CACI International Inc. – Virginia

May 2025 – Aug 2025

- Built a full-stack web app for **U.S. Customs field officers** using Angular, Spring Boot, and PostgreSQL.
- Expanded database test coverage **by 20%**, enhancing system stability and enabling more efficient bug detection.
- **Ramped quickly onto unfamiliar enterprise Spring Boot codebase** in Agile team, contributing through sprint planning, standups, and code reviews.

Data Science Research Intern, CDSS, UC Berkeley – Berkeley, CA

Jan 2025 – May 2025

- Fine-tuned BART transformer for Ancient Akkadian lemmatization in low-resource settings, **beating open-source baselines by 15%** using a custom edit-distance metric over lemma forms.
- Stepped up to lead disorganized team with no formal authority — **built shared task system**, assigned work by strength, held mock presentations, presented at UC Berkeley Discovery Symposium.

Software Engineering Intern, Floras – Berkeley, CA

Sep 2024 – Dec 2024

- Built a Flutter mobile app for carbon offset purchases, **integrating Patch.io** to display ESG data by region.
- Implemented **secure user authentication and token management**.

Software Engineering Intern, IT Hired Guns Inc. – Orlando, FL

Jan 2023 – Aug 2024

- Published **B2C cross-platform loyalty apps** (e.g., HotSpot, Kenjo) for unbranded gas stations using Flutter.
- Integrated location services and backend APIs to deliver **updates on nearby stations, fuel prices, and offers**.

Skills

Languages: Python, Java, JavaScript, SQL, Dart, Scheme

Technologies: React.js, Angular, Spring Boot, Flutter, Next.js, Firebase, MongoDB, Git, PyTorch, scikit-learn, Pandas, LangGraph, FastAPI, Playwright.